The relational model and normalisation



Data engineering · University · Lecture 1

In this lecture

- Use keys and dependencies.
- Normalise to 3NF and BCNF.
- Teach from anomalies.

Remember

- Dependencies decide the split.
- Lossless joins.
- Denormalise with a reason.

Break it, then fix it

1. Warm-up (5 min): Spot the problem
2. Break (15 min): Anomalies.
3. Derive (15 min): Dependencies.
4. Fix (20 min): Decompose.
5. Wrap-up (5 min): One-minute paper.

Let's discuss

1. Which anomaly appeared first?
2. Where did the dependencies come from?
3. When would you denormalise?

Exit ticket

- Write one functional dependency from a system you use.